Exam prep - this is the summary of all the content for the exam

Dynamic Modelling for Hum-Cent Sys (Vrije Universiteit Amsterdam)

Scannen om te openen op Studeersnel

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Exam prep

Chapter 1

Model - a representation of an object, system, or idea in some form other than that of entity itself

- A model is a simplification, not a complete representation of the reality
- A model is based on several choices and assumptions of the creator of the model
- Characteristics of the model are not

- a schematic description of a system, theory or phenomenon that accounts for its known or inferred properties and may be used for further study of its characteristic.

A dynamic model describes a specific phenomenon, theory system (i.e. a process); a dynamic model specifies changes over time

Elements of a dynamic model:

- Concepts
  - Represented as variables
- Timeline
  - Timepoint, stepsize
- Rules that describe how concepts change per time step
- Additional elements
  - Start values
  - Optional parameters
  - Optional stochastic variables

Modelling and simulation cycle

- There are **two main phases**:
  - Model **design** - the model is built
  - Model **analysis** - the model and its behaviour is studied
- The cycle consists of **four activities:**
  - **Conceptualization**: determining the main aspects and their relations
  - **Formalization**: specifying the detailed model
  - **Simulation**: performing experiments with the model to generate traces
  - **Evaluation**: verifying whether the model behaves as expected
- If the evaluation reveals that the model is not correct, the process starts again with a new conceptualisation phase
- Visualisation of simulations
  - Graphical representation of values of model variables at subsequent time points
  - Representation of truth values of model variables at subsequent time points (state trace)

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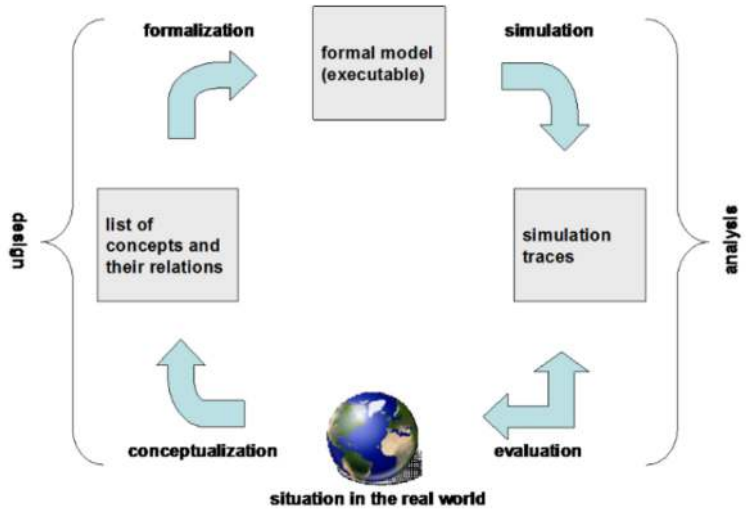

Designing a model

- Consists of conceptualization and formalization

Conceptualization

- Three types of tasks
  - Characterizing the process to be modelled
  - (Identification of sub-processes)
  - Concept identification
  - Relationship between concepts

Formalization

- Formalization needs to be done for both concepts and relations
- The formalization depends on the representation mechanism; two representations will be discussed:
  - Numerical representation
    - Concepts ate represented by variables with a value
    - Relations are rules that specify how the value of one variable can be calculated from the values of other variables
  - Logical representation
    - Concepts and their properties are represented by statements that can be true or false
    - Relations between concepts are rules that specify statements such that when at some time point they are true then another statement will be true of false

Analysis of and with a model

- Validation: checking whether the model is correctly built and whether it represents the reality in an adequate way

Validation

- Done by setting up and running simulation experiments and evaluating the resulting traces by comparing them to the expected outcome

Simulation

- The results of a simulation is a sequence of states of the model at subsequent time points which is called a "trace" or "simulation trace", or sometimes "simulation run"

Evaluation

- Consists of comparing simulation results with expectations

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- Expectations:
    - Based on reality (in case of validation)
    - Based on hypotheses (in case of studying the process)

Why modelling?

- Study a process to increase understanding:
    - Study the behaviour of a process over time
    - Performing "what if" simulations
- Basis for intelligent systems
    - Models are used to understand what is happening with humans (and what to do)
    - Models can make systems behave human-like

Use of models for scientific understanding

- Scientific method:
    - Research questions lead to hypotheses
    - Formulate specific and precise hypotheses:
    - Validate hypotheses on simulation traces

Model properties: an hypothesis about a model behaviour (i.e. the outcome of a simulation given some input and parameter value)

Advantages of modelling

- Processes are easier to study than in reality, because that is:
    - Expensive
    - Time consuming
    - Hard to measure
    - Hard to manipulate
    - Not ethical
    - …

Stochastic variables

- Parameters for which the values are determined by a probability
- With this kind of models, running the same model several times results in possibly different traces, which all reflect a different scenario for the same model

Chapter 2

Modelling "growth"

- Basic example of model of dynamic process
- At the same time: building block for models of other process in which growth plays a role
- Aim: generic model
- Inspiration from nature

Understanding growth

- Phenomenon growth:
    - Generic processes
        - Economy, biology
        - Relations, trust
- Inspiration: biological growth
    - Two causes of growth at cell level
        - Volume increase
        - Cell division

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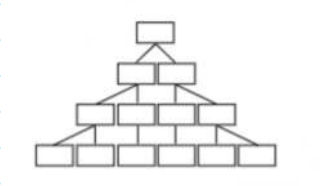
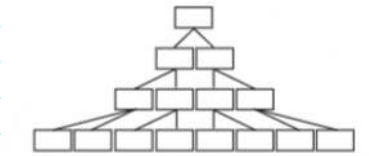
    - Two types of growth via cell division
        - Every cell divides
        - Only cells at the end divide

Generic growth patterns

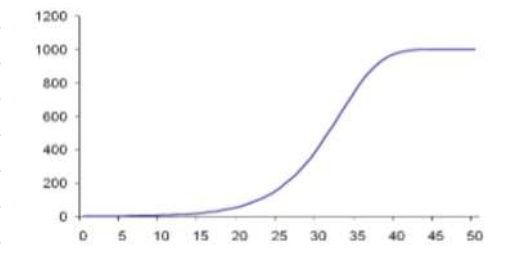
- Linear growth
    - Division by fixed number of cells
    - Equal gain per time step



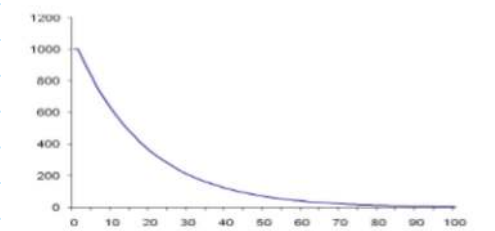
- Exponential growth
    - Division by every cell
    - Every time step doubling



- Limited growth (sigmoid or logistic growth)
    - Growth decreases when some maximum is approached
    - Growth is linear or exponential at start, then decreasing gain per time step



- Negative growth
    - Loss is larger than the gain
    - Exists in several variants
        - Linear
        - Exponential
        - …



Conceptualization of basic version

- Important concepts
    - Amount
    - Gain per time step
    - Loss per time step

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- Gain factor = gain as proportion of the total amount
- Loss factor = loss as proportion of the total amount

Formalisation

- Amount and gain factor affect gain

  Gain is calculated by multiplying the gain factor by the amount:

  $$Gain(t) = GainFactor(t)*Amount(t)$$

- Amount and loss factor affect loss

  Loss is calculated by multiplying the loss factor by the amount:

  $$Loss(t) = LossFactor(t)*Amount(t)$$

- From gain and loss to amount

  Gain adds to the amount and loss substracts from the amount, both taken over the time interval that is passing

  $$Amount(t+\Delta t) = Amount(t) + (Gain(t) - Loss(t)) * \Delta t$$

- Gain factor

  Constant factor for exponential growth

  $$GainFactor(t) = \alpha$$

Adapted formalization

- Limited gain:
  - Proportional to distance to maximum capacity (C)
    - When close to C, the gain is also small

      $$(C - Amount(t)) / C$$

  - Thus:

    $$GainFactor(t) = \alpha * ((C - Amount(t)) / C)$$

  - Results in "logistic" growth

Chapter 3

Modelling decision making

- Complex process
  - Much research about patterns
    - Psychological experiments
    - Decision models
  - Underlying mechanism are difficult to study
    - Brain imaging only shows activity of brain areas
  - Bacterium is simple enough to understand chemical pathways

Pattern

- Food intake behaviour:
  - At any point in time:

    If it observes that - at previous time point - glucose is present, then it will import glucose

    If it observes that - at previous time point - lactose is present, and that no glucose is present, the it will import lactose

Internal mechanism

- Bacterium can "observe" the environment (sensoring) and "decide: about
  - Lactose intake or

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    - Glucose intake
- Intake of lactose or glucose is controlled by two specific enzymes
- "lactose intake enzyme" is produced if:
    - Chemical substance CRPcAMP is present
    - (low) concentration of lactose in cell
- Substances in cell;
    - CRPcAMP is only present within cell if glucose is <0.1 mmol/l outside the cell
        - Indicator of absence of glucose
    - A little bit pf lactose is present in cell when there is lactose outside the cell (infiltrates)
- The bacterium will grow from both lactose intake as glucose intake

Conceptualization

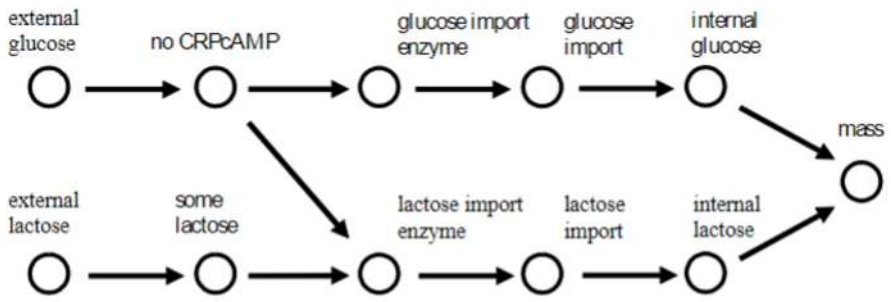
External state
- glucose externally present
- lactose externally present

Internal states reflecting the external state
- CRPcAMP internally absent (glucose indicator)
- some lactose internally present (lactose indicator)

Internal states for action initiation
- enzyme for glucose import present
- enzyme for lactose import present

Action performance
- glucose import takes place
- lactose import takes place

Internal state showing the effect of actions
- substantial glucose internally present
- substantial lactose internally present
- mass



Formalization of concepts

- glucose externally present: *EXTgluc* [0/1]
- lactose externally present: *EXTlac* [0/1]
- glucose indicator (no CRPcAMP): *glucIND* [0/1]
- lactose indicator: *lacIND* [0/1]
- glucose import enzyme: *glucimpENZ* [0/1]
- lactose import enzyme: *lacimpENZ* [0/1]
- glucose internally present: *INTgluc* [0/1]
- lactose internally present: *INTlac* [0/1]
- mass: *M* [real]

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Formalization of relations

- From external glucose to glucose indicator
  - $glucIND\ (t+\Delta t) = EXTgluc(t)$
- From lactose and glucose indicator to enzyme for lactose import
  - $lacimpENZ(t+\Delta t) = lacIND(t)*(1 - glucIND(t))$
- Effect on mass
  - $M(t+\Delta t) = M(t) + \gamma\,(INTgluc(t) + INTlac(t))*M(t)$

Chapter 4

Neural mechanism for learning

Behaviour before learning phase

- Initially the following behaviour is shown
  - A tail shock leads to a response
    (moving: contraction)
  - A light touch on its siphon is insufficient to trigger such a response

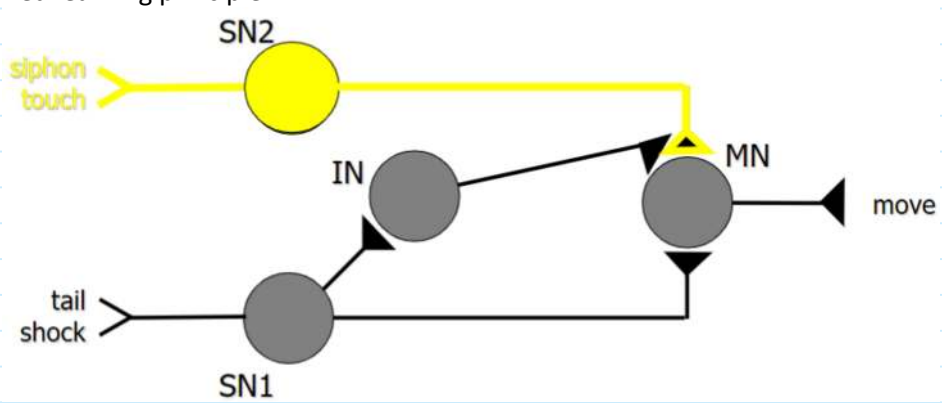
Behaviour after a learning phase

- It turns out that after a number of trials the behaviour has changed (**adapted** behaviour):
  - The animal also responds (moving) on the siphon touch

Training Aplysia

- Protocol for training an Aplysia:
  - A number of trials
  - In each trial:
    - The subject is touched lightly on its siphon
    - Then, shocked on its tail
    - As a consequence it responds

Simplified learning principle



- If both SN2 and In are activated simultaneously, this changes the synapse between SN2 and MN
- It makes that in this synapse more neurotransmitter is produced if SN2 is activated
- After a number of times this leads to the new situations that also activation of SN2 leads to activation of MN

Concepts

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tail shock (*TailShock*)
siphon touch (*SiphonTouch*)
contraction (*Contraction*)
sensory neuron SN1 is activated (*ASN1*)
sensory neuron SN2 is activated (*ASN2*)
motoneuron MN is activated (*AMN*)
the synapse S1 between SN1 and MN has a certain strength (*S1*)
the synapse S2 between SN2 and MN has a certain strength (*S2*)

Relations

Sensory processes

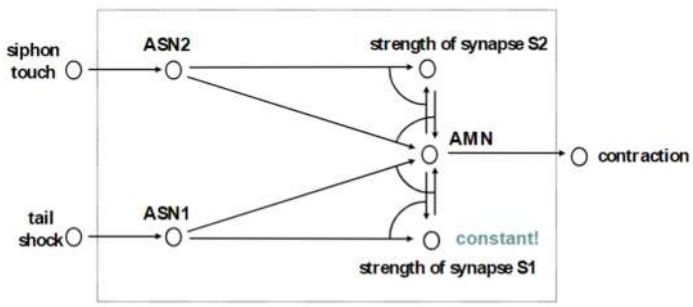
- A tail shock affects activation of SN1
- A siphon touch affects activation of SN2

Action generation

- Activation of SN1 affects activation of MN
- Activation of SN2 together with a certain strength of synapse S2 affects activation of MN
- Activation of MN affects contraction

Adaptation

- Simultaneous activation of MN and SN2 affects the strength of synapse S2 between SN2 and MN by increasing it



Formalization

- Concepts
  - Events and neuron activation: [0/1]
  - Synapse strenght: integer
- Relations
  - From tail shock to activation of SN1
    - $ASN1(t+\Delta t) = max(TailShock(t), TailShock(t-\Delta t), TailShock(t-2\Delta t))$
  - From siphon touch to activation of SN2
    - $ASN2(t+\Delta t) = max(SiphonTouch(t), SiphonTouch(t-\Delta t), SiphonTouch(t-2\Delta t))$
  - From activation of MN to contraction
    - $Contraction(t+\Delta t) = AMN(t)$
  - From activation of SN1 or SN2 to activation of MN
    - $AMN(t+\Delta t) = \begin{cases} 1 & \text{if } ASN1(t) = 1 \text{ or: } S2(t) = 4 \text{ and } ASN2(t) = 1 \\ 0 & \text{else} \end{cases}$
  - Adjustment of the synapse strength based on simultaneous activation of MN and SN2
    - $S2(t+\Delta t) = \begin{cases} S2(t) + 1 & \text{if } S2(t) < 4 \text{ and } AMN(t) = ASN2(t) = 1 \\ S2(t) & \text{else} \end{cases}$

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Chapter 5

Honeybee colony

- A society based on decentralized decision making
  - Individual decisions influence global processes
- Within honeybee colonies
  - Every worker bee has the same skills
  - There are a number of roles
  - Taking up a role is triggered by environmental cues

Roles

The forager

Tasks

- Get food from outside
- Store the food in the hive

Triggers to perform role

- Amount of food present

The brood carer

Tasks

- Get food available in the hive
- Feed hungry larvae with the food

Triggers to perform role

- Pheromones emitted by larvae

The undertaker

Tasks

- Remove corpses from the hive

Triggers to perform role

- Amount of corpses in the hive

The patroller

Tasks

- Patrols the hive
- Fights enemies it encounters in and around the hive

Triggers to perform role

- Enemies observed

Global aspects influencing individual behaviour

- Food availability
- Larvae strenght
  - Pheromone rate
- Cleanness
  - Dead bodies
- Safety
  - Enemies

Observations in honeybee colony

- Different bees take on different roles
- All together, the work is distributed in such a way that the most pressing tasks are performed by most bees
  - Doesn't run "out of hands"
- Some bees are taking up specific roles more often
  - Bees specialize

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Expected patterns of model

- Environmental parameters are kept between certain bounds by the individual decision making processes
    - Larvae strenght, safety, food storage, and cleanness
- Bees tend to keep their roles over longer periods
    - Specialisation of individuals is emerging in the process

Possible research question

- How long does it take to address sudden threats of the society' s parameters:
    - In how far are fast changing environments a problem

Cognitive level decision model

- Each individual maintains norms of each of the aspects
- For each aspect, the individual determines in how far the current state is unsatisfactory, expressed in a degree of urgency for that aspect
- For each aspect, the relative urgency is determines: the degree of urgency divided by the norm for that aspect
- The most urgent aspect is the one with highest relative urgency; the role is taken that relates to this aspect
- Norms are adapted: if an individual is in a role addressing a certain aspects, then the norm for that aspect is slowly decreased over tie

Simulation experiment

Validation of expected patterns:

- Are the aspects kept between bounds? Yes
- Does specialisation take place? Yes

Chapter 6

Use of virus simulation

- Understand the process of virus spread
- Predict future scenarios
    - Burden on hospitals
- Compare different intervention strategies

Our aim

- Create a model that can predict how virus spreads through society
- Possible research questions:
    - When infections transmit easily, will the whole population become infected?
    - Is it possible to adapt contact behaviour to avoid propagation through the whole population?
    - Is there a threshold for the amount of interaction such that the infection dies out?

Expected patterns?

- In the first phase, the amount of infected people increases almost exponentially
- At some point in time, the infection rate decreases
- Eventually, the number of recovered people is large enough to stop the spread

Modeling epidemics:

- Modeling individuals
    - Individual-based
- Modeling groups

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    - Population-based (collective)
    - Abstract

Principle of individual approach

- Individuals can be in different states:
    - Susceptible (not infected yet)
    - Infective, or
    - Recovered(immune and not infectious)
- Transmission of infection by contacts
    - When an infected and susceptible person meet
        - Contact frequency
        - Contact intensity

Concepts individual approach

- Infection state of a person
    - Susceptible
    - Infective
    - Recovered
- Location of a person
- Contact frequency per day
- Contact intensity
- Recovery rate
    - Chance per day for infective to recover

Individual level - formalisation

susceptible infective recovered	*InfectionStateA1 = 0, InfectionStateA2 = 0, ……* *InfectionStateA1 = 1 , …* *InfectionStateA1 = 2 , …*
location of individuals	*LocationA1 = 1, LocationA1 = 2, ……* *LocationA2 = 1, LocationA2 = 2, ……*
contact frequency per day (for one individual)	*ContactFrequency*
contact intensity	*ContactIntensity*
recovery rate (chance per day for infective to recover)	*RecoveryRate*

Concepts collective approach

- Same, but no individual infection
- Instead: different concepts for total number of people in a specific state
    - (the group of) susceptibles
    - (the group of) infected
    - (the group of) recovered

Collective level - formalisation

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number of susceptible individuals (non-infected)	*Susceptibles*
number of infective individuals	*Infectives*
number of recovered individuals (immune or dead, non-infective)	*Recovered*
number of relevant contacts per day (infective with susceptible)	*RelevantContacts*
number of infections taking place per day	*Infections*
contact frequency per person per day	*ContactFrequency*
contact intensity	*ContactIntensity*
recovery rate (fraction of infectives recovering per day)	*RecoveryRate*

Collective vs individual level

- Collective level
  - Sub groups represented by numbers
  - Abstraction from reality
  - Computationally efficient
- Individual level:
  - Each individual represented separately
  - More faithful
  - Computationally complex
  - Probabilistic element needed

Further explorations

- Compare the results of simulations by the model at the individual level and the model at the collective level
- Incorporate individual difference in contact behaviour
  - For example, consider a (core) population that has higher contact frequencies or intensities than other individuals
- Consider the case that recovered individuals become susceptible again:
  - No immunity after some time
- Consider the typical children's diseases, where all the time new births are adding individuals to the population
- Explore what can be done by vaccination programs
- Incorporate change of contact behaviour of individuals depending on the number of infectives
  - For example, if the number of infectives becomes higher, the contact frequency and/or contact intensity become lower
- Develop a model for transmission of an influenza infection from birds to humans
- Incorporate internal cognitive models for decision making about contact behaviour; for example, based on beliefs, desires and intention

Chapter 7

What is emotion contagion?

- Influencing others through conscious or unconscious induction of emotion states
- Tendency to mimic the nonverbal behaviour of others; synchronization of facial expressions, vocalizations, postures and movements with others, in order to "converge emotionally"

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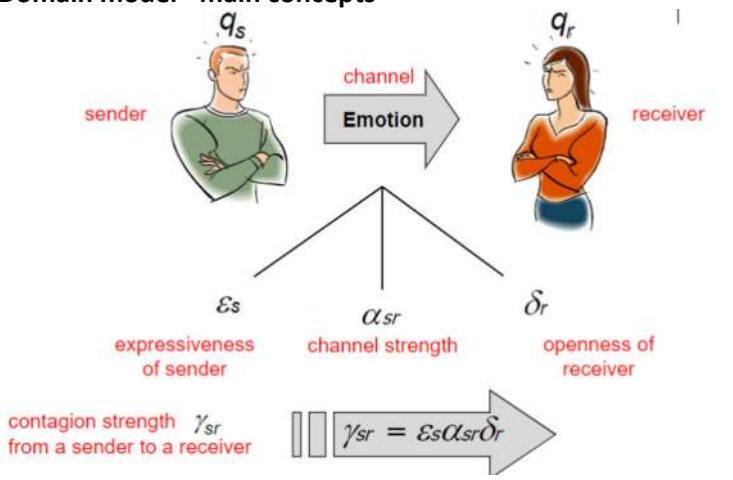
Domain model - main concepts



Variables

- the level of a person *A*'s emotion q_A

Parameters

- a person *A*'s emotion expressiveness ε_A
- a person *A*'s openness for received emotion δ_A
- the strength of the channel from sender *B* to receiver *A* α_{BA}
- the contagion strength from a sender *B* to a receiver *A* γ_{BA}

Note that some of the parameters can also be manipulated

Formalisation of concepts

- Predicates

Concept	Formalisation
person B has emotion expressiveness E	`has_expressiveness(B:AGENT, E:REAL)`
person A has openness for received emotion D	`has_openness(A:AGENT, D:REAL)`
channel from sender B to receiver A has strength C	`has_channel_strength( B:AGENT, A:AGENT, C:REAL)`
contagion strength from B to A is CS	`has_contagion_strength( B:AGENT, A:AGENT, CS:REAL)`
overall contagion strength to receiver A is CS	`has_overall_contagion_strength( A:AGENT, CS:REAL)`
person A has relevance factor R	`has_relevance(A:AGENT, R:REAL)`
person A has emotion level V	`has_emotion_level(A:AGENT, V:REAL)`
the group has emotion level V	`group_emotion_level(V:REAL)`

- Sorts

Sort	Description of use	Elements
AGENT	group members	a1, a2, a3, a, b, c, arnie, bernie, charlie
REAL	group member emotion level, group emotion level, expressiveness factor, channel strength, openness, contagion strength, relevance factor	real numbers

Detailed domain model - general idea

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At each time point, for each group member ***A***:

- Determine the **average emotion impact** q_A^* of the other group members on ***A***, taking into account that not all members have an equal impact on ***A***
- Determine the **difference** $q_A^* - q_A$ of this average q_A^* with the current emotion level q_A of person ***A***
- **Adjust** the emotion level q_A by a fraction γ of this difference, so that the new emotion level for ***A*** becomes $q_A + \gamma (q_A^* - q_A)$

- Another way of expressing this is that the new emotion level of ***A*** becomes a **weighted average** of the old emotion level q_A and the average value q_A^* of the other group members:
  $(1-\gamma)\, q_A + \gamma\, q_A^*$

Chapter 8

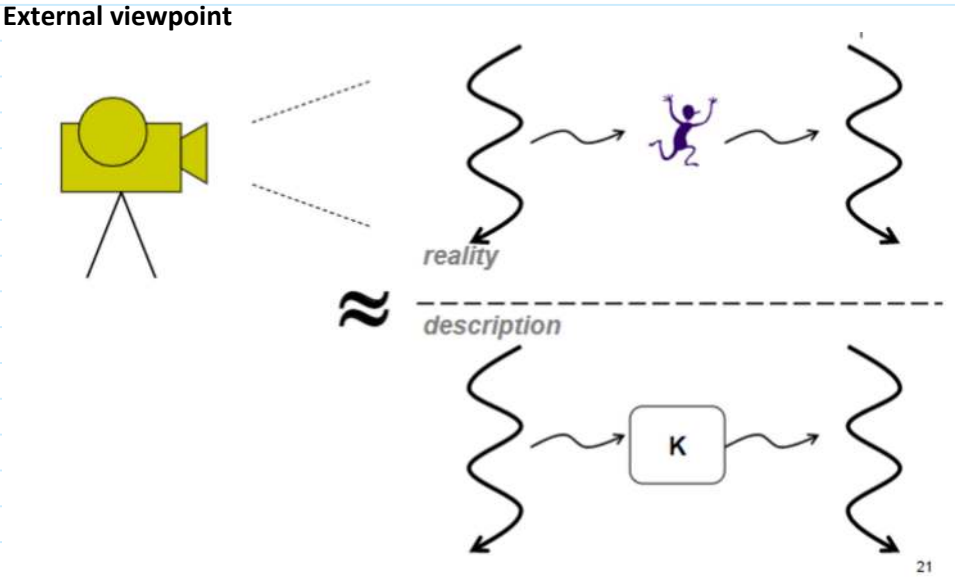
What is behaviour?

- Behaviour is a **range of actions and mannerism** made by organisms, systems or artificial entities **in conjunction with their environment**, while includes the other systems or organism around as well as the physical environment
- It is the **response** of the system or organism to various stimuli or **inputs**, whether **internal or external**, **conscious or subconscious**, **overt or convert**, **and voluntary or involuntary**

Two viewpoints on modelling world knowledge

- External (observers) viewpoint
  - Knowledge exploited as a useful means to externally explain the subject's behaviour without making any claim on the actual internal functioning of the subject
- Internal (the subject's own) viewpoint
  - A subject actually has knowledge, encoded in one way or the other in it (cognition), and this knowledge is used by the subject to generate its behaviour

External viewpoint



Internal viewpoint

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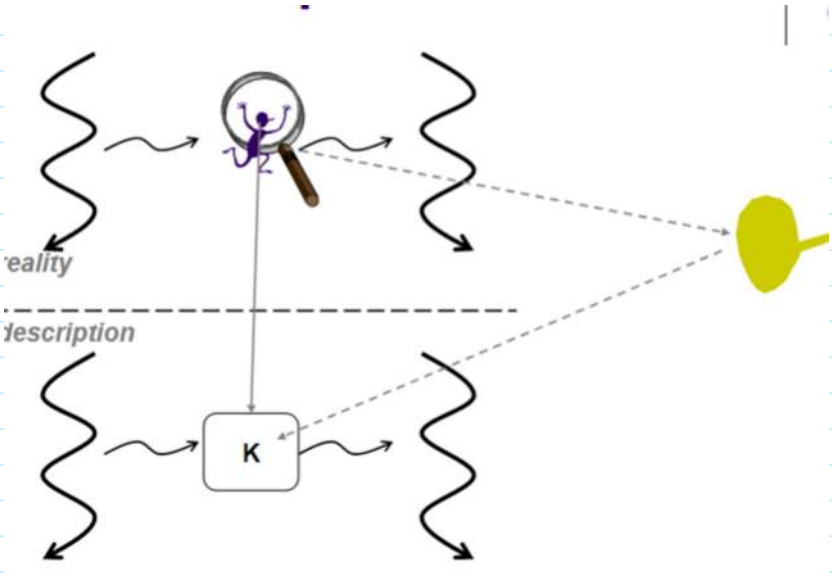

Central theme in cognitive science

- Analyse how cognitions relate to behaviour (or : how internal descriptions relate to external ones)
- Approach: modelling and simulation
    - External descriptions are the "characteristics patterns" that are used to provide evidence for the feasibility of internal descriptions

Environmental complexity thesis

- The function of cognition is to enable the agent to deal with environmental complexity
- The more complex the environment, the more sophisticated the behaviour that is required to deal with this environment
- The more sophisticated the behaviour that is required to cope with this environment the more complex the mental representations and capabilities needed

Research questions

From an **external** behavioural perspective:

- Which concepts relating to input and output of an individual are needed to **describe** the pattern of behaviour?
- How can the dynamic relationship between input and output be described using these concepts?

From an **internal** cognitive perspective:

- What types of **internal cognitive states** are required to generate the behavioural pattern?
- How can the dynamic relationships between these internal cognitive states be describes?

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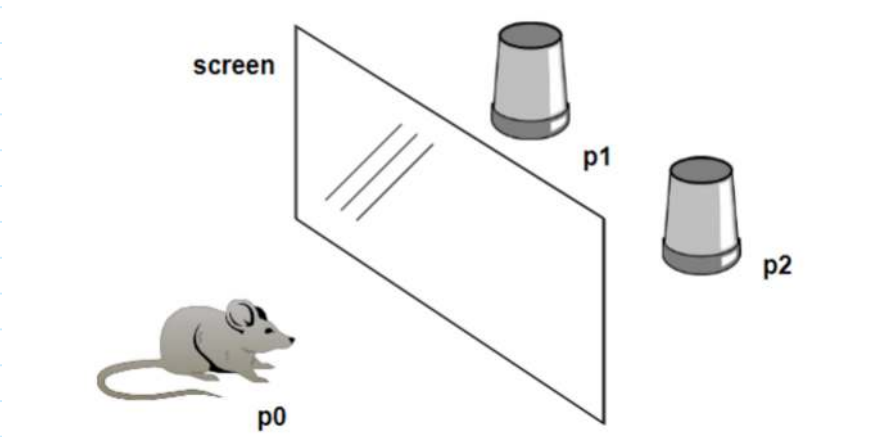
	concepts describing states	dynamic relationships describing processes
external behavioural view	*input and output state properties* - observed stimuli, incoming communication - actions, outgoing communication	*temporal relationships involving input and output state properties* - from observed stimuli and incoming communications to actions and outgoing communications
internal cognitive view	*cognitive state properties* - sensory representations, preparations - beliefs, desires, intentions - sensitivity levels	*temporal relationships involving cognitive state properties* - from sensory representations to preparations - from experiences to sensitivity levels - from sensitivity level to action - from desires to intentions - from intentions to actions 27

Chapter 9

Experimental setup

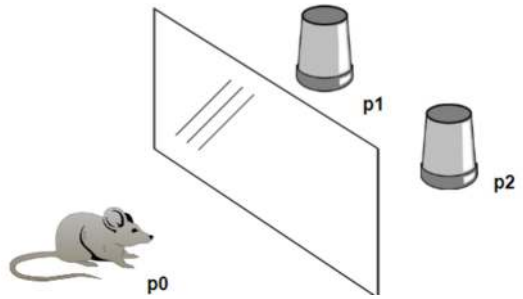


Situations

1. Empty cup at p1 and p2
2. Empty cup at p1,
   Food (visible) at p2
3. Empty cup at p1,
   Food at p2,
   Cup at p2 (food no longer visible)
4. As 3. but food was never visible

Results: situation 1

- Animal shows no preference for p1 or p2
- Animal may wander or stay at p0



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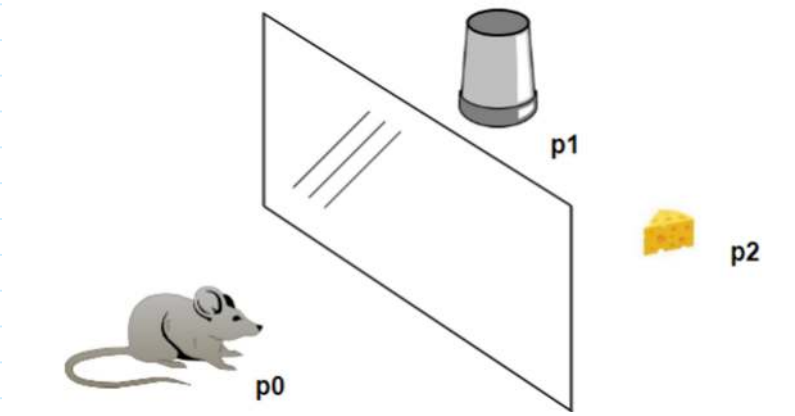
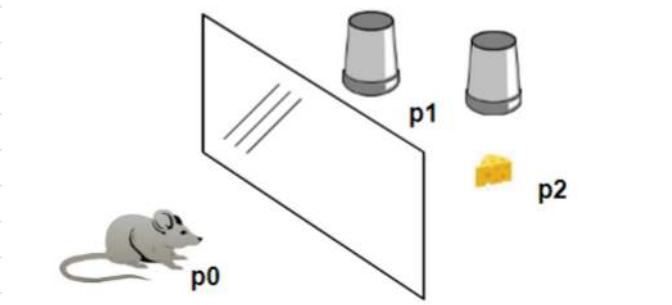
Results: simulation 2

- Animal goes to p2
  (explanation: **stimulus-response behaviour**)



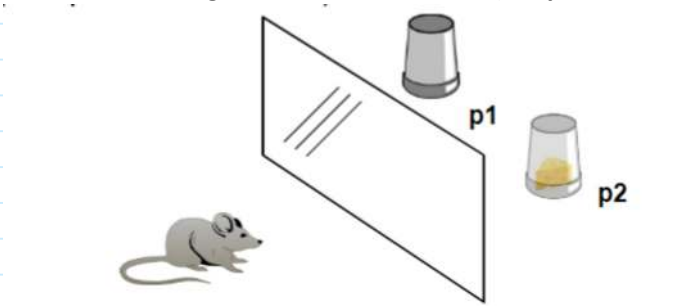
Results: situation 3

- Different possibilities:
  - Some animals behave as in situation 1
  - Some animals go to p1
    (explanation: **delayed response behaviour**)



Results: situation 4

- Most animals behave as in situation1, but:
  - Some animals may search food (**motivation-based behaviour**)
  - Some animals might find food, and learn (**adaptive behaviour**)



Types of behaviour

- Type A: **stimulus-response,** no delayed response
- Type B **: delayed response**
- Type C: **motivation-based**
  - Behaviour depends on motivation or goal
  - May act pro-actively In situation 1 (hungry)
  - May ignore food stimuli in situation 2

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- Type D: **adaptive**

Reactive behaviour

- Includes the following two types
  - **Stimulus-response behaviour**
  - **Delayed response behaviour**

Stimulus-response behaviour: example traces

time *trace*	*0*	*1*	*2*	*3*	*4*	*5*
trace 1	food at p2 screen	no food screen	food at p2 screen	food at p2 no screen	food at p2 no screen goes to p2	no food no screen goes to p2
trace 2	no food screen	no food screen	no food screen	food at p2 screen	food at p2 no screen	food at p2 no screen goes to p2
trace 3	no food screen	no food no screen	no food no screen	food at p2 no screen	food at p2 no screen goes to p2	no food no screen goes to p2
trace 4	food at p2 screen	no food no screen	food at p2 screen	food at p2 screen	food at p2 no screen	food at p2 no screen goes to p2

Describing behaviour in logical way

- By means of executable dynamic properties
  - Logical relations that specify truth value dependencies between state properties at next time step
- During conceptualisation we use a semi-formal notation
  - FOR all TIMEPOINTS:
    IF…THEN…(at next time step)

Description of Behaviour

- From an external perspective

FOR all TIMEPOINTS
IF the animal observes food at p2
AND the animal observes
that there is no screen
THEN the animal will go to p2
(at the next time point)

Modelling the behaviour

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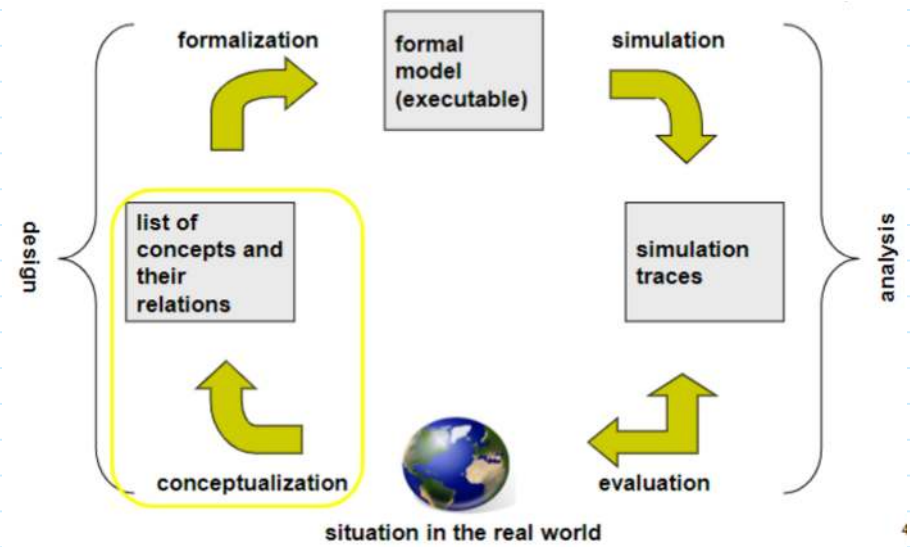


Conceptualization

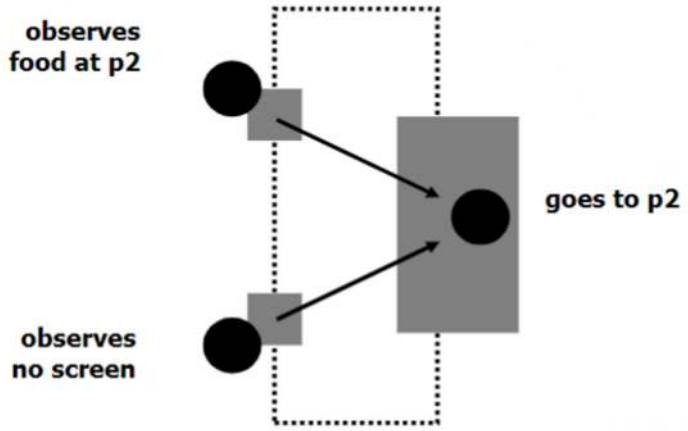
- Input: observation
  (of presence of food and screen)
- Output: action
  (of going to p2)

Relations between Concepts

- Stimulus-response behaviour

An observation that there is food at p2
affects
that it will go to p2.

An observation that no screen is present
affects
that it will go to p2.



Internal viewpoint

- What would be needed from an internal viewpoint?
  - Specification of internal cognitive states
- Internal: sensory representations

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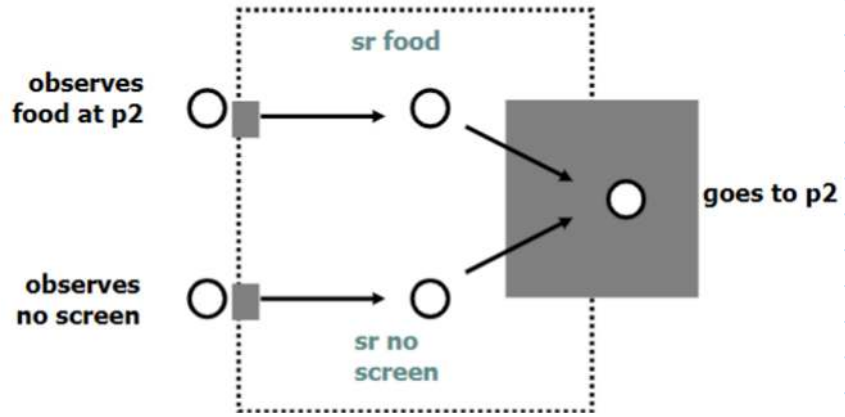
(of presence of food and screen)



Sr = sensory representation

Delayed response behaviour

- A more sophisticated form of reactive behaviour

Delayed response behaviour: example traces

	0	*1*	*2*	*3*	*4*	*5*
trace 1	food screen	cup screen	cup screen	cup no screen	cup no screen **goes to p2**	cup no screen **goes to p2**
trace 2	no food screen	no food screen	cup screen	cup screen	cup no screen	cup no screen
trace 3	no food screen	no food no screen	no food no screen	no food no screen	food no screen	food no screen **goes to p2**
trace 4	food screen	no food screen	food screen	cup screen	cup no screen	cup no screen **goes to p2**

Delayed response behaviour: description

at any point in time,

if the animal observes that no screen is present

and at some earlier point in time the animal observed that food was present at position p2,

then the animal will go to position p2

Delayed response behaviour: alternative description

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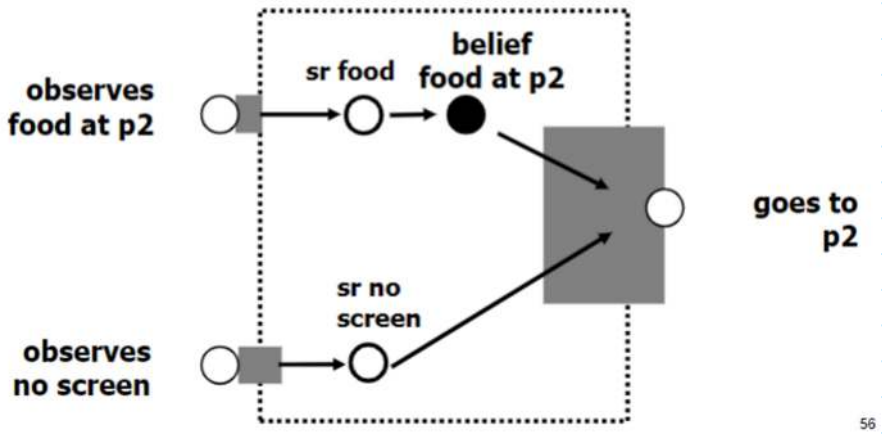
at any point in time t1,
if the animal observes that no screen is present
and at some earlier point in time t it observed that food was present at position p2,
and at every point in time t' after t up to t1, it did not observe that no food was present at p2,
then the animal will go to position p2

Internal cognitive states

- How do we represent this at an internal cognitive level?
- Internal: belief
  (of presence of food and screen)

Relationships between concepts



Formalisation of the concepts

External behavioural level		State properties
observation	(of presence of food) (of presence of screen)	observed(at_position(food, p2)) observed(screen)
action	(goto p2)	performed(goto(p2))
Internal cognitive level		
sensory representation	(of presence of food) (of presence of screen)	sr(at_position(food, p2)) sr(screen)
belief	(of presence of food) (of presence of screen)	belief(at_position(food, p2)) belief(screen)

Formalizing logical relations

- Remember: **executable dynamic properties**
  - Logical relations that specify truth value dependencies between state properties at next time step

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Warning: facts always need to be derived explicitly

- No closed-world assumption
  - If no truth value is generated for a predicate, there is no information about state in next time step
- Therefore**, two types of dynamic properties**:

Step Property (new facts in next state)

If X holds

then Y will hold

$$X \twoheadrightarrow Y$$

Persistence property (maintaining facts in next state)

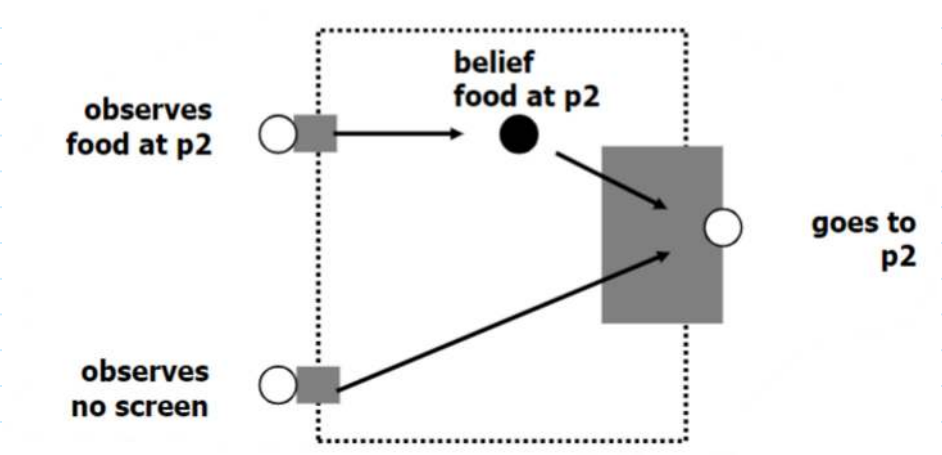
If X holds

(and Y does not hold)

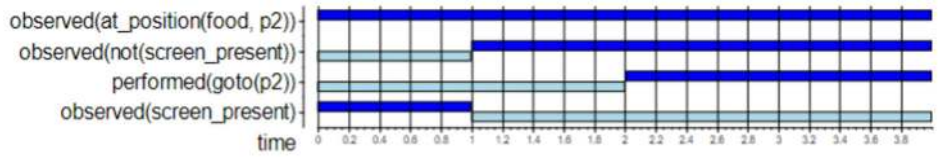
then X will hold

$$X \wedge \neg Y \twoheadrightarrow X$$

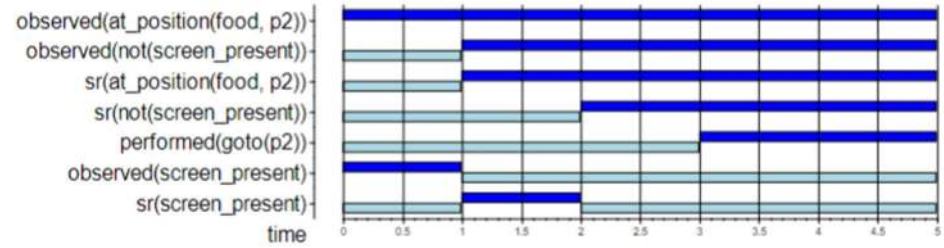
Simplified version of the internal dynamic of delayed response behaviour



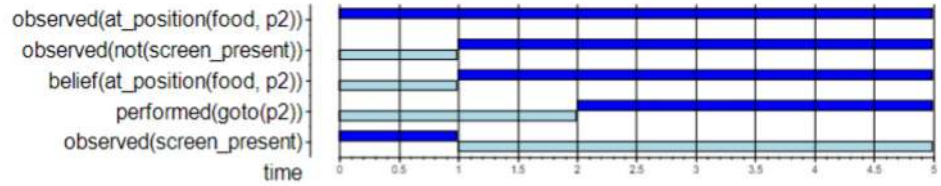
Stimulus response behaviour (external) simulation trace



Stimulus response behaviour (internal) simulation trace



Delayed response behaviour (internal) simulation trace



Take home message

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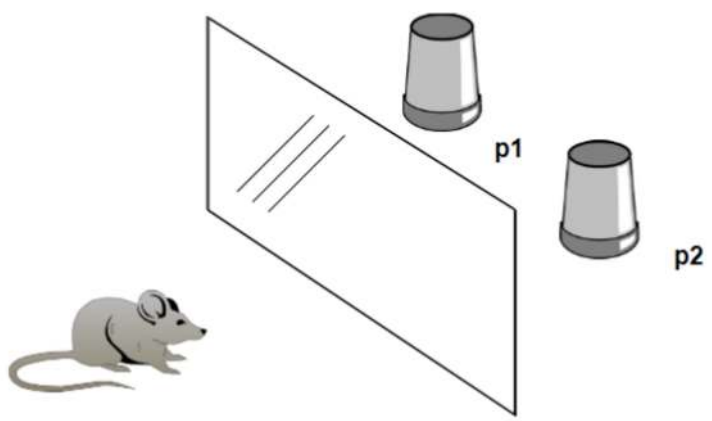
- Modelling and simulation of behaviour requires observations and actions
- Types of behaviour:
  - Stimulus -response
  - Delayed response
  - Motivation-based
  - Adaptive
- External vs internal viewpoint
  - External viewpoint is not always possible to model
    - Complex behaviour requires complex mental states!
- Logical formalism is useful representation

Chapter 10

Motivation-Based behaviour

- Patterns of behaviour in which the animal
  - Takes the initiative
    (without observable direct reason)
- Viewed from external and internal viewpoint

Experimental setting



Motivation based behaviour: example trace

	time 0	*time 1*	*time 2*	*time 3*	*time 4*	*time 5*
trace 1	food screen	no food no screen	food screen	food screen	food no screen	food no screen **goto p2**
trace 2	no food no screen	food no screen	no food screen	no food no screen	no food no screen	no food no screen
trace 3	no food no screen	no food no screen	food screen	food no screen	food no screen	food no screen
trace 4	no food no screen	food no screen	no food screen	food screen	food no screen	food no screen **goto p2**

External explanation of motivation-based behaviour

- Why does the animal go to p1?

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- The animal starts wanting food if it cannot get it; it then reacts immediately if food is present and screen is removed

The animal goes to p2, because
it just observed the absence of the screen, and at an earlier point in time
it observed food at p2,
whereas at a still earlier point in time
it observed the absence of food at p2, and
at yet another still earlier point in time
it observed the presence of food at p2.

Motivation-based behaviour: external description

at any time point t1
if the animal observes the absence of the screen,
and there exists time points s1, s2, and s3
with $s1 < s2 < s3 \leq t1$ such that
at s3 the animal observed the presence of food at p2,
and at s2 the animal observed the absence of food at p2,
and at s1 the animal observed the presence of food at p2,
then after time point t1 the animal will go to p2

External explanation of motivation-based behaviour

- More a historical type of explanation than based on a direct cause
- Doubtful whether this is a **satisfactory** explanation
- For an explanation based on a direct causes, internal state properties seem to be required

Conceptualization: specification of input and output

- Input: observation
  (of presence of food and screen)
- Output: action
  (of going to p2)

Specification of internal cognitive states

- Internal: belief
  (on presence of food and screen)

  desire
  (to have food)

  intention
  (to go to p2)

BDI model (belief-desire-intention)

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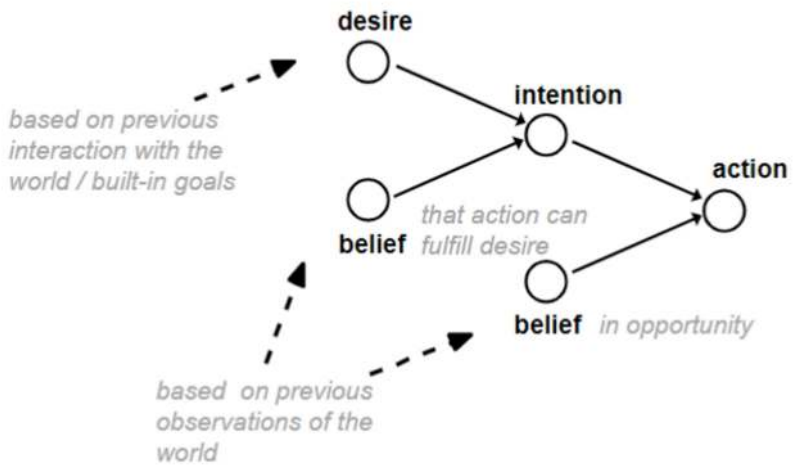
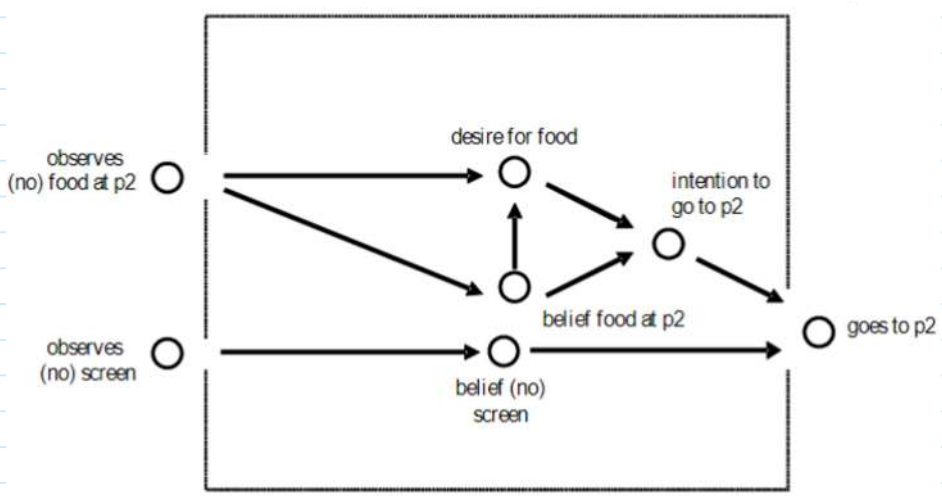


Concepts

- observes_food_at_p2
- observes_no_food_at_p2
- observes_no_screen
- belief_no_screen
- belief_food_at_p2
- desire_for_food
- intention_to_go_to_p2
- goes_to_p2

Relationships between concepts



Example simulation traces

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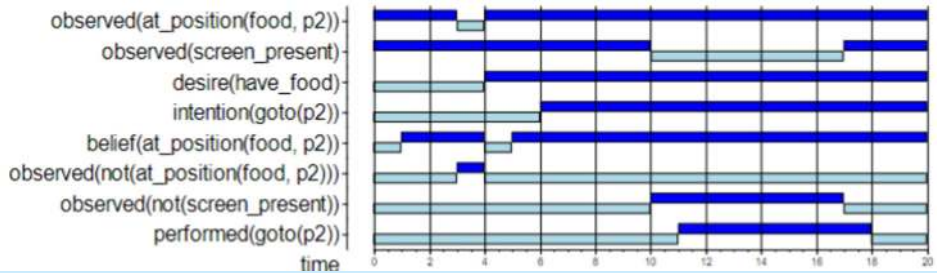
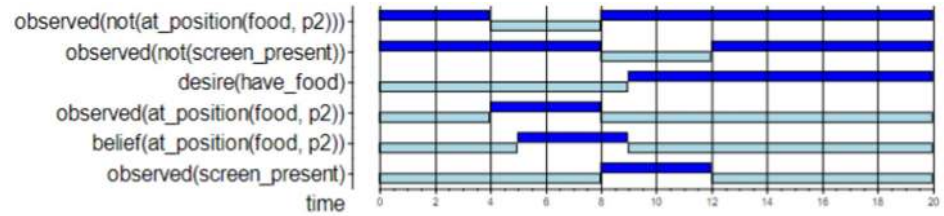
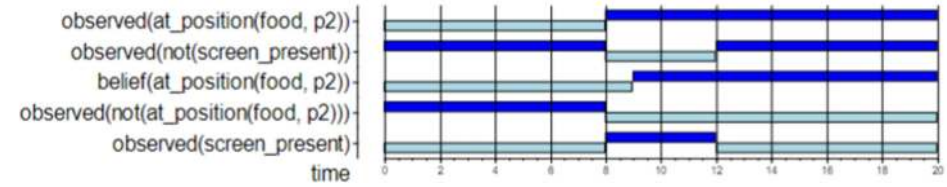


- No **intention** generated



- No **desire** generated



Motivation-based behaviour: summary

- Patterns of behaviour in which the animal takes the initiative
- External perspective: may require nontrivial references to past input and/or output
- Internal perspective: using internal state properties such as beliefs, desires, intentions

What are the internal states?

- Two perspectives:
  - Realism
    - The internal cognitive state properties are (known to be) related to certain "real" physical (e.g. neurological) state properties
  - Instrumentalism
    - The internal cognitive state properties are not (known to be) related to "real" physical state properties. They are only instruments, e.g. to do predictions on behaviour

Intentional stance

A specific instance of an instrumentalist view is Dennett's intentional stance:

  - Attribution of intentional state properties to others
  - Intentional strategy to explain and predict behaviour

- Treat the other as a rational individual
- Figure out what beliefs it has, given its place in the world and its purpose
- Figure out what desires it ought to have, given its place in the world and its purpose
- Predict that the individual will behave according to its desires and belief

Physical stance

- This can be opposed to the physical stance (realist view):
  - Determine the physical constitution of a system
  - Use knowledge of laws of physics to predict the behaviour depending on the inputs
  - Often useful for engineers

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Advantages of intentional stance

'Predicting that someone will duck if you throw a brick at him is easy from the folk-psychological stance; it is and will always be intractable if you have to trace the photons from brick to eyeball, the neurotransmitters from optic nerve to motor nerve, and so forth.'
(Dennett, 1991), p. 42

- An intentional stance description abstracts from specific aspects of physical realisation; therefore
  - The description is more concise and transparent
  - It is more generic: it applies to more cases, in different physical realisations

Conclusion

- Humans often try to understand and influence others by reasoning about other human's beliefs, desires and intentions
  - This is crucial for functioning in human society
- Intentional stance is philosophical basis for intelligent systems
  - If we want to build systems that "understand" people, it is useful to model their behaviour from an intentional stance

Chapter 11

Adaptive behaviour

- Patterns of behaviour in which the individual after some point in time shows different behaviour than before, due to, for example
  - Learning
  - Trauma
  - Forgetting
- This new behaviour is persistent over a longer time period

Environmental complexity thesis

- **Stimulus - response** behaviour
  - Works only with visible food
- **Delayed response** behaviour
  - Food becoming invisible is taken into account
- **Motivation-based** behaviour
  - Food becoming absent is taking into account
  - Never visible food is taken into account
- **Adaptive behaviour**
  - Accommodate behaviour to environment

Wasp example: before

- Dynamic relationship satisfied by traces before the adaptation:

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for all time points until some time point t
if the individual observes that a wasp **is close** to the individual,
then the individual will **not panic**

Wasp example: after

- Dynamic relationship satisfied by traces after the adaptation (adapted behaviour):

for all time points after some time point t
if the individual observes that a wasp **is close** to the individual,
then the individual will **panic**

Wasp example: the process

- What happened to you that makes you behave in this way?
- Why are you the way you are?
  - "behaviour after" compared to "behaviour before" shows the result of adaptation but does not show the adaptation process
- Which dynamic relationships characterise the adaptation process itself?

Example trace

	0	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*
trace 1	wasp close	wasp close	wasp stings	wasp close ouch!	wasp close panic	no wasp panic	no wasp	wasp close	wasp close panic
trace 2	wasp close	wasp close	no wasp	no wasp	wasp close	wasp close	no wasp	wasp close	wasp close
trace 3	wasp close	wasp stings	no wasp ouch!	no wasp	wasp close	wasp close panic	no wasp panic	wasp close	wasp close panic
trace 4	wasp close	wasp close	no wasp	wasp close	wasp stings	wasp close ouch!	no wasp panic	wasp close	wasp close panic

External explanation of adaptive behaviour

Suppose a trace is given

	0	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*
trace 4	wasp close	wasp close	no wasp	wasp close	wasp stings	wasp close ouch!	no wasp panic	wasp close	wasp close panic

Why does the individual panic at time point 8?

The individual panics because at time point 7 is a wasp close

Specification of input and output

- Input: **Observation** of stimuli
  (of a wasp being close)
  Experience

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(of being stung)

- Output: **action**
  (of: panicking, screaming "ouch", doing nothing)

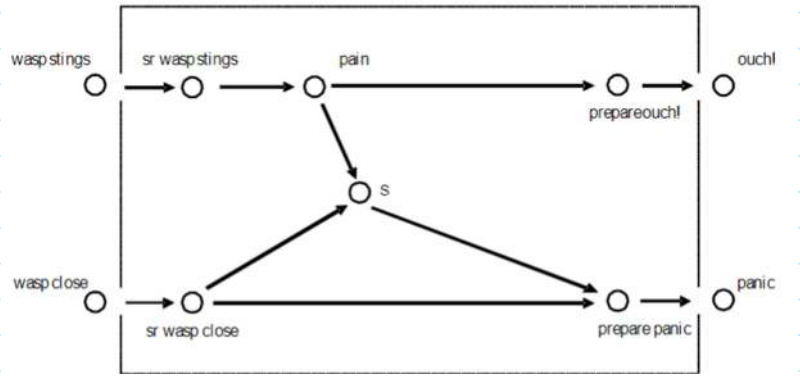
Specification of internal cognitive states

- Internal: **sensory representation**
  (of stimuli and experiences)

  Sensitivity
  (for wasps)

  Preparations
  (for actions)

Relationships between concepts



Formalisation of the concepts:

External behavioural level	
observation of stimuli of experiences	 (wasp close) (being stung)
action	(panic) (scream ouch!)
Internal cognitive level	
sensory representation	(of wasp close) (of being stung)
sensitivity	(for wasp)
preparation	(for panicking) (for screaming ouch!)

Example simulation trace

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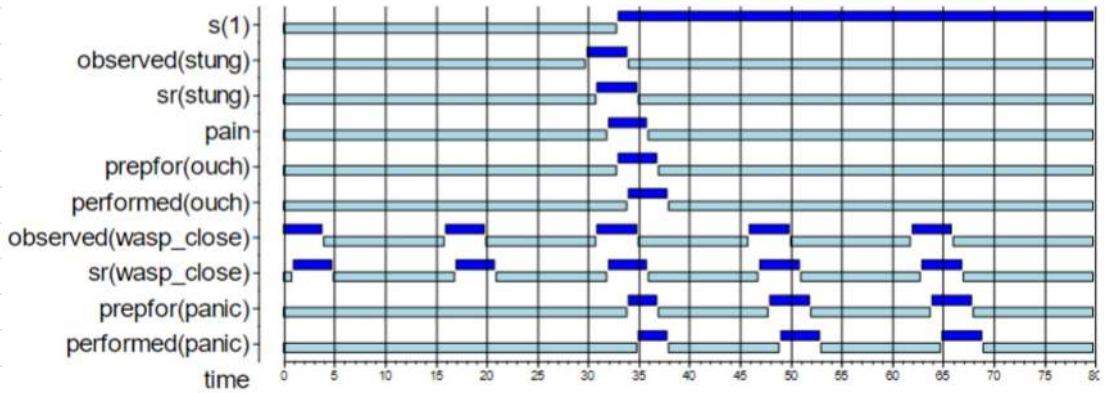

Gradual and relative adaptive behaviour

- Sensitivity increases due to a number of sting events
- The more often a wasp has stung the individual, the more panic the individual will show, when a wasp is close

Chapter 12

Mind extended in the world

Environmental complexity thesis

- **Stimulus - response** behaviour
  - Works only with visible food
- **Delayed response** behaviour
  - Food becoming invisible is taken into account
- **Motivation-based** behaviour
  - Food becoming absent is taking into account
  - Never visible food is taken into account
- **Adaptive behaviour**
  - Accommodate behaviour to environment
- **Extended mind behaviour**
  - Environment participates in the behaviour

Extended mind in the world

- Extended mind state for an individual:
  - A physical state in the external world used by the individual in its mental processes
  - Created by an action of the individual
  - Observed by the individual

Extended mind: examples

- **Little thumb**
  - While walking, put pebbles on the ground regularly
  - To find the way back, follow the pebbles
- **Making lists**
  - While having something to be done in mind, or something desired, write down on the proper list
  - When free look at the list to see what to do

Extended mind: characteristics

- Environment participates in individual's mental processes
- Individual's **mental processes** are **simplified**
- More intensive **interaction** with the world

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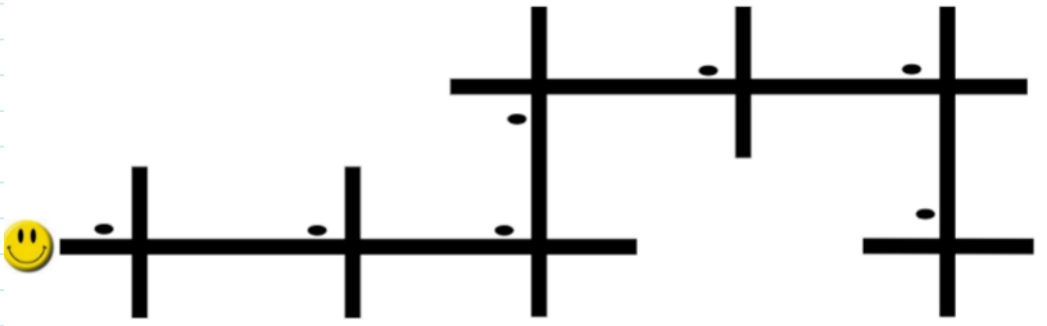
- Individual **depends** on external world:
  - Delegation of some mental representations
  - Delegation of some mental capabilities
  - Il Little Thumb's pebbles are lost, he's lost

Case study: little thumb

- Drops a pebble before each crossing
- On way back, he follows the pebbles



Specification of input and output

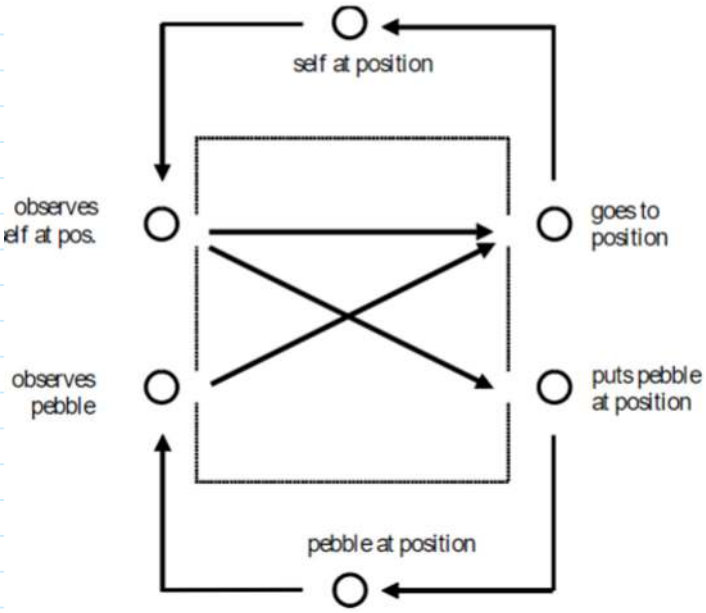
- Input: **observation**
  (of crossing ahead)
  (of a pebble at a certain position)

- Output: **action**
  (drop a pebble)
  (go to a position)
  (change direction)

Specification of External world states

- External: **world fact**
  (of a pebble at a certain position)

Relationships between concepts



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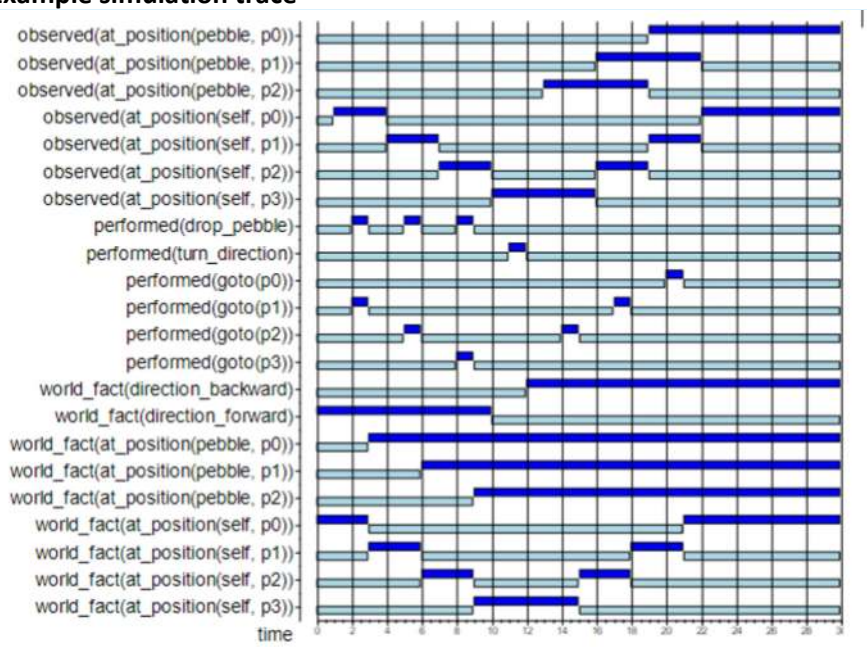
Detailed model: formalisation of the concepts

observation	(of position of self) (of position of pebble)	observed(at_position(self, P)) observed(at_position(pebble, P))
action performance	(drop pebble) (go to position) (change direction)	performed(drop_pebble) performed(goto(P)) performed(turn_direction)
pebble at a position in external world self at a position in external world adjacent positions in external world direction forward direction backward		world_fact(at_position(pebble, P)) world_fact(at_position(self, P)) world_fact(adjacent(P1, P2)) world_fact(direction_forward) world_fact(direction_backward)

Example simulation trace



Evaluation

- Assumptions made:
  - Pebbles and their positions persist
  - No other pebbles are there than the ones put by little thumb
  - Pebbles are always found back by observation
  - The way back is clearly distinguished from the way forth. For example, on the way forth no pebbles are followed, and on the way back no pebbles are dropped

Further exploration

- Bread crumbs instead of pebbles
- Shared extended mind by multiple individuals
- Behaviour based on "to-do lists"
- Pheromone-based behaviour of ants

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